

Racial Disparities in Mortgage Lending: A 2023 HMDA Replication of the Boston Fed Study

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1. INTRODUCTION

Access to a mortgage is one of the key determinants of homeownership and the overall outcomes of the housing market. Because most people cannot pay-in-full for a real estate property, and thus must rely on mortgage financing, the mortgage application process is a significant barrier to entry into the housing market. Economists have long deliberated upon whether racial bias unnecessarily increases this barrier to entry. A study published by researchers at the Federal Reserve Bank of Boston in 1996 found substantial racial disparities in mortgage denial rates even after controlling for relevant financial characteristics of borrowers. This paper aims to examine whether those results still persist using the 2023 Home Mortgage Disclosure Act (HMDA) data for the Boston metropolitan statistical area.

The findings of the Boston Fed study were published in *Mortgage Lending in Boston: Interpreting HMDA Data* by Munnell et al. (1996). According to the authors, the HMDA data for 1990 showed significantly higher denial rates for Black and Hispanic applicants than for White applicants and “[i]n fact, high-income minorities in Boston were more likely to be turned down than low-income whites” (Munnell et al., 1996). Because the HMDA data lacked crucial underwriting variables considered by lending institutions in the mortgage application process, the Federal Reserve Bank of Boston conducted a supplemental survey requesting additional applicant and loan information from lenders. The analysts found that with the additional survey information, even after controlling for financial, employment, and neighborhood characteristics, Black or Hispanic mortgage applicants in the Boston metropolitan area were approximately 56 percent more likely to be denied than comparable White applicants. These results sparked significant debate regarding the existence of racial bias in mortgage lending.

Since the initial findings were published, replications of this analysis have been conducted multiple times, most recently and notably by researchers at the Federal Reserve Bank of Philadelphia in 2024. According to the authors of this study, the motivation behind their replication was to “take advantage of expanded confidential HMDA data to provide the first estimates of contemporary disparate treatment that are comparable to the seminal Munnell et al. (1996) work from the 1990s” (Bhutta et al., 2024). Additionally, the researchers focus much of their study on the evolving differences in the mortgage lending process when compared to the

1990s, particularly in the adoption of Automated Underwriting Systems (AUS). The present analysis shares the same motivation of reassessing potential racial disparities in the modern mortgage lending market. While the Philadelphia Fed analysis utilizes a confidential HMDA dataset covering 2018 and 2019, adopts a national scope, and incorporates additional information unavailable in the public HMDA extract, this paper instead focuses on publicly available post-COVID HMDA data within the Boston MSA, thereby providing a geographically targeted update that aligns with the setting of the original study.

To accomplish this objective, this study constructs a Boston-area applicant sample and estimates mortgage denial models that mirror the structure of the 1996 analysis while incorporating modern specifications. These specifications include comparable sample restrictions, borrower-characteristic controls, and lender fixed effects to account for institutional heterogeneity, thereby maintaining the replication-oriented design. However, to better reflect the evolving demographic makeup of Boston, the present study's models incorporate more contemporary racial and ethnic definitions. Additional models that control for tract differences, AUS usage and model, and examine the probability of denial for high income earners are included to assess the stability of the results of the main models. Together, these modeling choices allow the analysis to balance accurate replication of the original study while adjusting to contemporary institutional and demographic conditions, ensuring an effective evaluation of any persistent racial disparities in the modern Boston mortgage loan market.

The results indicate that after controlling for loan, borrower, and lender attributes, racial disparities remain for mortgage loan applicants in Boston in 2023. In particular, one model estimates an absolute denial gap of 2.4 percentage points between minority applicants and comparable White applicants, roughly a 46 percent increase in the predicted probability of denial. Another model incorporating lender fixed effects suggests that an absolute denial gap of approximately 2.1 percentage points exists between all minority applicants and comparable White applicants within the same lending institutions, roughly a 35 percent increase in predicted denial probability. The robustness models and evidence from prior literature display similar findings. Black applicants appear to account for a majority of the disparities, as demonstrated by this study's granular race model. However, it should be noted that differences in data availability,

model specifications, and institutional conditions prevent direct comparisons regarding changes in racial disparities over time. Furthermore, the sources of these denial gaps, whether attributable to racial bias or alternative factors, cannot be determined from the results of these models. What can be established is that in 2023, even after controlling for relevant observable characteristics, measurable racial denial gaps exist for minority mortgage applicants in Boston despite substantial changes to underwriting practices over the past three decades.

This paper proceeds in five sections. Firstly, the methodology and findings of the original Boston Fed study and the underwriting process are outlined in Section 2. Next, the public extract and confidential post-2018 HMDA data, sample restrictions, and outlier handling is discussed in Section 3. Section 4 establishes the variable generation, racial and ethnic definitions, and logistic regression methodology utilized in the analysis. Findings of the main and robustness models are subsequently presented and explored in Section 5. Finally, the interpretation of the analysis is summarized and concluded in Section 6. Supplementary visualizations, including summary statistics, regression results, predicted loan denial probability figures, and marginal effect estimates, are provided throughout the text and in the Appendix.

2. BACKGROUND

Due to this paper’s focus on replication, it is important to understand the context and methodology of the study it aims to expand upon. The Home Mortgage Disclosure Act (HMDA), enacted in 1975, requires many financial institutions to collect and publicly disclose loan-level mortgage data, providing transparency for policy creation and enabling analysts to detect potential discriminatory lending patterns (Consumer Financial Protection Bureau, 2019b). In 1991, researchers at the Federal Reserve Bank of Boston identified that higher rates of denial for minorities relative to White applicants were evident in the recently published 1990 HMDA data. According to Munnell et al. (1996), these findings initiated debate about the origins of these differences, with some believing “that these disparities in denial rates are evidence of discrimination on the part of lending institutions”, while others “including lenders, argue that such conclusions are unwarranted because the HMDA data do not include information on credit histories, loan-to-value ratios, and other factors considered in making mortgage decisions”.

Although the 1989 amendments to the HMDA required lenders to add sex, race, income, and loan outcome information for potential borrowers, the lack of key applicant and loan characteristics in the 1990 data ultimately prompted the researchers' further analysis.

To supplement the limited HMDA data, the Federal Reserve Bank of Boston, with help from other supervisory agencies, surveyed financial institutions in the Boston MSA. This survey provided information on financial and employment variables that lenders deemed pertinent to mortgage approval (Munnell et al., 1996). According to the authors, the combination of Census neighborhood information and these 1990 data, which were requested for all Black or Hispanic conventional mortgage loan applicants and a random sample of 3300 White applicants, allowed for an in-depth examination of approval outcomes. Surveyed institutions were limited "to those that had received at least 25 mortgage applications from borrowers of all races" (Munnell et al., 1992). Additionally, individuals who were refinancing, applying for nonresidential properties, lacked data for any key explanatory variables, or withdrew their application were excluded. These restrictions helped ensure the sample captured a comparable population of mortgage applicants while maintaining a clean dataset for interpretable econometric modeling.

Using the combined HMDA data, Census tract attributes, and survey results, the researchers could proceed with their analysis. Since lending institutions aim to maximize their expected profit, they prioritize minimizing the likelihood and costs of default for each mortgage. Therefore, the researchers posited in their original 1992 working paper that the probability that a lender denies an applicant, $P(D)$, could be written as a function of multiple loan and applicant factors. That is,

$$P(D) = f(F, R, L, T, C)$$

where F represents the applicant's ability to carry the loan, R represents the risk of default, L denotes the potential losses the institution would incur in the event of default or foreclosure, T reflects the loan terms, and C represents the potential influence of bias, whether due to race or other personal characteristics, that may sway the lender's judgement (Munnell et al., 1992). The dataset variables were then organized into these categorical factors and analyzed using regression

analysis. Logistic and ordinary least squares regression were used for the prediction models, the mechanisms of which will be further discussed in Section 4.

As previously stated, the researchers found that for a Black or Hispanic individual with average White economic attributes, their probability of denial increases by 56 percent (Munnell et al., 1996). Nearly all of the thirteen additional robustness models demonstrated the same statistically significant and positive racial coefficient, even after controlling for more in-depth variables, including those relating to financial health, tract characteristics, and lender-specific effects. However, the authors emphasized the limited scope of these findings, as the observed disparities may originate from other sources of discrimination unrelated to the underwriting process. Munnell et al. (1996) believed that a lesser quality of education or differential treatment in the labor market, which could systematically affect the economic profiles of minority applicants, may have contributed to the inequality. While the analysts were unable to definitively identify the root cause of their results, they ascertained that a measurable denial gap was evident.

Understanding the mortgage application process and the changes it has undergone since the 1990s provides valuable background. At the time of the original study, being approved for a mortgage involved three main steps: a review of the viability of the application, verification of applicant information and property appraisal, and evaluation of the relevant and potential “compensating” factors (Munnell et al., 1992). Firstly, a prospective borrower would go to a convenient lending institution and fill out a loan application form. Based on the information provided on the form, the underwriter made an initial assessment of the applicant’s viability. If the loan officer deemed the application to be subpar, the applicant could be immediately denied and the process ended there. If, instead, the loan officer felt the applicant had a legitimate chance of being approved, then the underwriter attempted to verify the provided information with evidence that the applicant was able and willing to repay the loan. This was determined with proof of current and continued employment, bank deposits, and a credit history report.

Once verified, financial indicators, such as the applicant’s monthly expense-to-income ratio and debt-to-income (DTI) ratio, were scrutinized. Such ratios played a substantial role in determining the lending institution’s ability to sell the loan on the secondary market, largely to government-sponsored enterprises (GSEs) like Fannie Mae and Freddie Mac (Munnell et al.,

1992). The GSEs proposed guideline thresholds for these ratios, which tended to be roughly 30 percent, but lenders were not required to adhere to them strictly. Next, the lending institution would conduct a property appraisal and calculate the LTV ratio, for which purchasers on the secondary market typically maintained an 80 percent benchmark. If the underwriter remained uncertain of the security of the loan, ‘compensating factors’, such as “a large down payment, ... , a strong propensity to save, ... , and an excellent potential for future income” (Munnell et al., 1992), could be considered to offset other concerning attributes. Finally, the loan officer was prepared to approve or deny the applicant for a mortgage loan.

Much of this process remains true today with a few important differences. Firstly, the rise of the internet allows the mortgage application process to be largely, if not entirely, completed online (Bhutta et al., 2024). The second, and most notable, difference is the modern adoption of automated underwriting systems, which analyze the applicant’s information and provide a recommendation on the approval or denial of the application. The models that the researchers at the Philadelphia Fed focus their study on are Fannie Mae’s Desktop Underwriter (DU), Freddie Mac’s Loan Product Advisor (LPA), and the Federal Housing Administration’s credit risk scorecard (TOTAL). Crucially, these AUS take relevant economic factors into account like any loan officer would, but are not provided any racial information. Since the AUS do not make the ultimate lending decision but instead provide recommendations to underwriters, Bhutta et al. placed significant emphasis on examining whether differences between AUS recommendations and the underwriter’s final conclusion varied systematically. The analysts find that although racial denial gaps persist in AUS assessments after controlling for HMDA factors, this disparity largely disappears once unobserved risk characteristics and lender strictness are accounted for (Bhutta et al., 2024).

3. DATA

This analysis is conducted using the 2023 public HMDA extract for the Boston MSA. The decision to use the 2023 dataset was made to ensure complete applicant information, as it is the most recently finalized reporting year at the time of this study. Additionally, this dataset reflects post-COVID mortgage applications and outcomes, allowing for an updated assessment

relative to studies that rely on pre-pandemic data. Since its inception, the HMDA data have undergone numerous amendments resulting from congressional bills. The Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010, which notably added credit score to the dataset, and the Economic Growth, Regulatory Relief, and Consumer Protection Act of 2017 are two such bills (Consumer Financial Protection Bureau, 2019a). Furthermore, the Consumer Financial Protection Bureau (CFPB) passed the 2015 HMDA rule, which added new or revised existing data points, importantly introducing debt-to-income ratio and automated underwriting systems identifiers. These revisions, which began to take effect in 2018, permitted researchers to conduct deeper analysis using key underwriting variables that were previously absent from the HMDA data.

However, the public HMDA extract has its limitations. To prevent the identification of individual applicants, variables such as credit score and AUS decision are omitted from the public data but remain available in the confidential extract. As a result, the model specifications rely on borrower and loan characteristics to estimate risk profiles, which will be discussed in Section 4. While the selected controls reflect important aspects of the underwriting process, they do not fully replace credit scores or automated underwriting decisions. Even researchers who have access to the confidential HMDA extract do not solely rely on it to examine racial bias due to its lack of information on factors where discrimination could arise in the mortgage application process. For instance, the analysts at the Federal Reserve Bank of Philadelphia utilize the National Mortgage Database (NMDb) and the National Survey of Mortgage Originations (NMSO) to explore “if lenders differentially discourage minorities from even submitting a formal application” and “whether lenders exhibit bias in their measurement of applicant income” (Bhutta et al., 2024). Importantly, the purpose of the present study is not to identify the sources of racial disparities but rather to assess their existence, for which the public HMDA extract is well suited.

The raw 2023 Boston MSA public HMDA extract contains information on 17,959 applicants. Several restrictions relating to loan, property, and applicant characteristics were imposed to maintain the replication-orientated focus of this analysis. First, the sample is limited to individuals purchasing single-family properties with one to four units as their principal

residence. The analysis further restricts the data to conventional, first-lien mortgage applications that are not open-end lines of credit, reverse mortgages, or for business or commercial purposes. Applicants whose institutions were exempt from reporting requirements or who lacked key underwriting variables, including DTI ratio, LTV ratio, property value, income, or loan amount, were excluded. Observations with missing, indeterminate, or joint race or ethnicity classifications were also removed to allow for clear racial comparisons. Finally, applicants reporting negative income values were excluded and observations above the 99th percentile of income, loan amount, and property value were trimmed to minimize the effect of outliers. These restrictions were implemented to align with the methodology of the original Munnell et al. study and to effectively assess differences within the Boston homebuyer population.

The final restricted sample retains 9,107 observations. While this total is smaller than that of the original dataset, the resulting sample size is sufficient to conduct an interpretable analysis. Table 1 below presents the sample distribution by race and ethnicity, along with the corresponding percentages and denial rates. The table indicates notable differences in denial outcomes across groups, with minority applicants experiencing higher denial rates than White applicants, which is consistent with Munnell et al.'s findings. This sample distribution demonstrates the demographic changes Boston has experienced over the past three decades. According to Munnell et al. (1992), the 1990 Census reported that 85 percent of Boston residents were White, 6.8 percent were Black, 4.5 percent were Hispanic, and 3.7 percent were classified as some other racial designation, including all Asians. In contrast, the most recent Census estimates that non-Hispanic Whites account for 44.1 percent of Boston's population, while Black residents make up 20.5 percent, Hispanics or Latinos 19.3 percent, and Asians 10.4 percent (U.S. Census Bureau, 2025). The racial distribution of the applicant sample differs from the overall population of Boston as reported by the Census, due to the fact that the HMDA captures mortgage applicants rather than all residents. However, the final sample adequately reflects the racial and ethnic developments of the city, which is considered when defining variables and will be further discussed in Section 4.

Summary statistics by race and ethnicity, including median income, mean debt-to-income ratio, and mean loan-to-value ratio, can be referenced below in Table 2. While Munnell et al.

Table 1: Sample Distribution and Denial Rates by Race and Ethnicity

	Observations	Share	Denial Rate
2 or more minority races	21	0.23	19.04
American Indian or Alaska Native	16	0.18	12.50
Asian	1396	15.33	7.31
Black	778	8.54	14.91
Native Hawaiian or Pacific Islander	15	0.16	6.67
White	6430	70.61	4.43
Hispanic	451	4.95	8.87
Total	9107	100.00	6.04

Share and Denial Rate reported as percentages; Share represents each group's proportion of the sample; Denial Rate represents the percentage of applications denied within each group.

Table 2: Summary Statistics by Race and Ethnicity

	Income	DTI Ratio	LTV Ratio
2 or more minority races	121 (47.87)	42.83 (3.94)	88.02 (9.70)
American Indian or Alaska Native	135 (90.95)	41.36 (3.91)	81.31 (16.77)
Asian	144 (126.48)	42.88 (4.02)	74.36 (16.06)
Black	116 (90.61)	43.12 (3.71)	87.58 (12.61)
Native Hawaiian or Pacific Islander	155 (125.86)	42.18 (3.92)	85.14 (16.25)
White	174 (139.45)	42.21 (3.99)	75.18 (17.94)
Hispanic	125 (102.91)	43.38 (3.77)	84.79 (14.16)
Total	157 (134.42)	42.49 (3.97)	76.65 (17.54)

Median income in 000s of USD; mean values for DTI and LTV Ratios; standard deviations in parentheses.

(1992) documented the historical adherence to a 30 percent debt-to-income ratio threshold, the 2023 sample demonstrates shifts in this standard. On average, debt-to-income ratios exceed the 1996 benchmark for all races, which coincides with higher debt normalization under the CFPB's Qualified Mortgage (QM) rule. The rule established general underwriting criteria requiring "that the consumer have a total (or "back-end") debt-to-income ratio that is less than or equal to 43 percent" (Consumer Financial Protection Bureau, 2013). The Bureau emphasizes that this is a general guideline, and that applicants who exceed this threshold can still be approved with the lending institution's discretion. The findings presented in Table 2 show that contemporary applicants in Boston tend to cluster around this benchmark.

Additionally, the average loan-to-value ratios for several racial groups surpass the previously established 80 percent criterion. However, as documented by Munnell et al. (1992), secondary markets would accept loans above this threshold contingent on the borrower's purchase of private mortgage insurance. As a result, applicants, particularly those belonging to minority groups, were approved despite higher LTV ratios, as reflected in the median loan-to-value ratio of 85 percent for approved Black or Hispanic applicants in the 1996 sample. The corresponding average LTV values of the present sample, 87.58 percent for Black applicants and 84.79 percent for Hispanic applicants, are comparable to the Munnell et al. figure and may reflect the continuity of flexible underwriting practices. Median income differences also remain present across racial and ethnic groups, with White borrowers earning significantly more on average.

4. METHODOLOGY

This section outlines the generation and definition of key variables, in addition to an overview of the modeling techniques and specifications used to estimate racial disparities in mortgage loan outcomes. The primary explanatory variable in the analysis is a binary minority indicator derived from the race and ethnicity fields reported in the HMDA data. The variable is equal to one if an applicant identifies as a racial or ethnic minority and is zero otherwise, with non-Hispanic White borrowers serving as the omitted reference category. Defining a minority as any individual who does not identify as non-Hispanic White represents one of the key departures

from the methodology of the Boston Fed researchers. As previously discussed, the largest minority populations in Boston in 1996 consisted of Black and Hispanic residents, with all other minorities comprising only 3.7 percent of residents. Furthermore, non-Black and non-Hispanic minority borrowers represented only 17 of 4,153 applicants in the supplemental survey sample collected by the analysts, prompting the authors to exclude these individuals from the final sample entirely. Thus, to focus on estimating racial disparities for the largest minority groups in the population, Munnell et al. (1996) chose to define their minority indicator to be equal to one for applicants identifying as Black or Hispanic and zero otherwise, with White applicants serving as the omitted reference category.

Given the substantial racial and ethnic changes Boston has experienced, the original variable definition would not be suitable for this study. Non-Black and non-Hispanic minorities now make up a significantly larger share of the city's residents and excluding them from this paper's final sample would fail to capture the full extent of potential disparate treatment affecting the minority population as a whole. For example, Asian applicants, who were excluded by Munnell et al., constitute the largest minority group in the present sample, exceeding the combined total of Black and Hispanic borrowers. Thus, the minority indicator was defined to reflect these contemporary demographic conditions.

Nevertheless, this definition implicitly assumes a level of homogeneity across racial groups, which may conceal meaningful differences. Considerable variation exists both within and across demographic categories, as suggested by the extreme standard deviations for income and LTV ratio reported in Table 2. To account for such variability, a more granular race model is estimated. Racial groups consisting of relatively few applicants, namely those identifying as two or more minority races, American Indian or Alaska Native, or Native Hawaiian or Pacific Islander, are aggregated into a "Non-Primary Minority" dummy variable. Consolidating these groups in such a manner aims to reduce sparsity and improve the precision of the maximum likelihood estimates. However, the relatively small number of observations in this category warrants the cautious interpretation of the corresponding regression coefficient, which will be expanded upon in Section 5. Additionally, White applicants are defined as individuals identifying as White and not Hispanic. Hispanic applicants are defined as those identifying as both White

and Hispanic. Black applicants include all individuals identifying as Black or African American. Asian borrowers are defined as any applicants identifying as Asian.

Several additional borrower and underwriting variables require clarification. To reflect standard underwriting thresholds, applicants' debt-to-income ratios were discretized into four categories: less than 36 percent, greater than or equal to 36 percent and less than 43 percent, greater than or equal to 43 percent and less than 50 percent, and greater than or equal to 50 percent. Established in the *Fannie Mae Selling Guide*, a debt-to-income ratio of 36 percent represents the upper bound generally accepted for manually underwritten loan applications (Fannie Mae, 2026). However, Fannie Mae (2026) explains that this bound can be extended to 45 percent if the borrower meets certain credit score and financial asset requirements. Additionally, as touched upon in Section 3, the CFPB's Qualified Mortgage rule recommends a 43 percent threshold for DTI ratios (Consumer Financial Protection Bureau, 2013). Finally, 50 percent represents the upper limit commonly used by Fannie Mae and Freddie Mac for loans analyzed through their respective automated underwriting systems, Desktop Underwriter and Loan Product Advisor (Federal Housing Finance Agency Office of Inspector General, 2019).

The automated underwriting systems indicators also deserve explanation. The evaluation models used by each applicant's lender during the underwriting process were grouped into four categories: GSE AUS Used, Internal AUS Used, Other AUS Used, and Exempt/NA AUS Usage. GSE AUS Used captures individuals who were evaluated using Fannie Mae's DU or Freddie Mac's LPA. Internal AUS Used reflects applicants analyzed using a lender's proprietary automated underwriting system. Other AUS Used catches applications ran through the Federal Housing Administration's (FHA) TOTAL Scorecard, the U.S. Department of Agriculture's Guaranteed Underwriting System (GUS), or some other model. Applicants whose lenders were exempt from reporting AUS usage or did not report a system were placed into the Exempt/NA AUS Usage category. Importantly, the indicator variables equal one if the corresponding automated underwriting system type appears in any of the HMDA AUS reporting fields. Because lenders may report up to five systems for a single application, these indicators are not mutually exclusive.

Income and loan-to-value ratio are kept as continuous variables in all model specifications rather than being clustered into categorical indicators. Being continuous changes the interpretation of the estimated coefficients of these variables compared to those of categorical dummies, which will be further elaborated upon later in this section. Since these data points take on purely numerical values in the HMDA data, unlike DTI ratio, which includes both numerical and interval figures, keeping them as continuous variables was convenient for the analysis. However, income was not directly included in any of the final specifications and was instead entered using its natural logarithmic form. Because only applicants who reported a positive income were kept in the final sample, the logged variable is well-defined for all observations.

The choice to use this regressor reflects the diminishing marginal effect of additional income on loan approval probability. A fixed percentage increase in income is expected to have a similar effect across applicants, whereas the effect of an equivalent increase in absolute income will vary depending on the individual's initial income. For example, increasing the salary of an individual earning \$100,000 by \$50,000, which represents a large proportional change, imposes a significantly greater effect on the probability of approval than providing the same \$50,000 increase to an applicant already earning \$350,000, due to a relatively smaller proportional change. These variables comprise the primary set of controls included in the regression specifications used to estimate racial differences in mortgage denial outcomes.

An overview of the mechanism of the primary modeling technique clarifies the methodological decisions underlying this study. Regression analysis is a statistical tool that quantifies the relationship between one or more exogenous, or independent, variables and an endogenous, or dependent, variable. This association can be interpreted by evaluating the estimated coefficients corresponding with each predictor relative to the model specification. By assigning specific values to the included independent variables, the model can predict the outcome of the dependent variable, which in this study is the probability of mortgage loan denial. Geometrically, the goal of regression is to generate a curve, or higher dimensional manifold, of best-fit that minimizes the effect of random noise while maximizing the model's forecasting ability.

An ordinary least squares (OLS) regression model assumes a linear relationship between the included exogenous variables and an endogenous variable. The equation estimating this relationship is generated by calculating the slope and intercept parameters that minimize the sum of squared residuals. OLS estimation is best suited for models predicting the outcomes of continuous dependent variables. The coefficient of determination, R^2 , measures the quality of this fit by representing the model's explained sum of squares (ESS) as a proportion of the total sum of squares (TSS). Because R^2 is derived from squared deviations relative to a fixed mean, the measure inherently assumes linearity. Central to OLS are the assumptions that, for every possible value the exogenous variables take on, the error terms have an expected value of zero, are homoskedastic, and are uncorrelated with the independent variables and each other. However, when applied to a binary dependent variable, such as loan denial, the assumptions of OLS are regularly violated and its predictive ability is reduced, necessitating a modeling technique that is more appropriate for discrete endogenous outcomes.

One such framework is logistic, or logit, regression. This model assumes the existence of an unobserved latent variable, z , that captures the underlying propensity for a binary outcome to occur and is modeled as a linear combination of the explanatory variables. A discrete outcome is realized only when the latent variable crosses a certain threshold. It is conventional to standardize this threshold to be zero, thereby allowing the model to be more intuitively interpreted. In this context, positive coefficients indicate an increase in the latent propensity, and thus probability, of the occurrence of the event, while negative coefficients represent a decrease. Therefore, for observation i , the probability that event one occurs can be modeled using the logistic cumulative distribution function (CDF), $F(z)$, as a result of n explanatory regressors. That is,

$$P(Y_i = 1 \mid X_{i,1}, \dots, X_{i,n}) = F(\hat{\beta}_0 + \hat{\beta}_1 X_{i,1} + \dots + \hat{\beta}_n X_{i,n})$$

where

$$F(z) = \frac{1}{1 + e^{-z}}, \quad F : \mathbb{R} \rightarrow (0,1).$$

The inherent confinement of the logistic CDF's codomain to the interval (0,1) makes it superior to OLS for modeling probabilities.

Unlike OLS, where coefficients represent the constant marginal effect a regressor imposes upon the dependent variable, interpreting the absolute values of the coefficients in a logit estimation is slightly less clear. Isolating for z in the logistic CDF results with

$$P = \frac{1}{1 + e^{-z}} \implies z = \ln\left(\frac{P}{1 - P}\right)$$

where, as previously established

$$z = \hat{\beta}_0 + \hat{\beta}_1 X_{i,1} + \cdots + \hat{\beta}_n X_{i,n}.$$

Thus, the estimated slope term, $\hat{\beta}_k$, denotes the constant change in the log-odds of an outcome associated with a one-unit increase in its corresponding regressor, X_k , for all $k \in \{1, \dots, n\}$. For indicator variables, this change is interpreted relative to the omitted reference category. The intercept, $\hat{\beta}_0$, reflects the log-odds of the outcome when all continuous regressors equal zero and all categorical variables take their omitted reference categories. These statistics are derived using maximum likelihood estimation (MLE), which selects the estimates that maximize the likelihood of observing the sample. The model's fit is measured using Pseudo- R^2 , which compares the log-likelihood of the estimated model with that of a null model, rather than measuring the proportion of variance explain by the linear R^2 . According to the developer of the metric, an excellent fit is represented by a Psuedo- R^2 that lies between 0.2 and 0.4 (McFadden, 1977).

Because log-odds are less directly interpretable than linear effects, average marginal effects (AMEs) can be calculated to illustrate the average change in the dependent outcome associated with a one-unit change in a regressor, leaving all other variables unchanged. For an indicator variable, this is accomplished by evaluating the logistic CDF for each observation using its observed regressor values. The predicted probability of the outcome is calculated first with the indicator variable set equal to one and then recalculated with the variable set equal to zero,

holding all else equal. The difference in estimated probability for each observation is computed and these differences are averaged for all observations, thus generating the average marginal effect of that regressor. Averaging the predicted probabilities for all observations directly without taking the differences produces the model's predicted probability of the outcome for a given categorical group.

Having outlined the variable construction and estimation framework, the model specifications can now be summarized. Every model aims to predict the probability of loan denial for a mortgage applicant in Boston in 2023. Standard errors are clustered by lender to account for potential correlation in unobserved factors influencing applications being considered by the same institution. Model 1 represents this study's baseline specification and controls for minority status, income, loan-to-value ratio, and debt-to-income ratio. Model 2 uses a more granular racial definition to examine differences across demographic groups while retaining the same non-racial regressors as the baseline. Model 3 adds lender fixed effects, adopting the approach of Munnell et al. (1996) and Bhutta et al. (2024), to control for institutional variation and within-lender racial disparities. Models 1-3 represent the main specifications.

Model 4 restricts the sample to high-income applicants earning above \$150,000 annually, approximately the sample's median income, to assess whether disparities persist among wealthier borrowers. Model 5 controls for neighborhood differences with the specification's inclusion of Tract-to-MSA Income Percentage. Model 6 evaluates whether AUS usage impacts differences in denial rates and Model 7 accounts for the specific automated underwriting system used by the institution. Models 4-7 serve as the robustness specifications that aim to validate the consistency of the findings of the main regressions.

5. RESULTS

Tables 3 and 4 report the results for Models 1–3. Table 3 presents the regression estimates, while Table 4 reports the corresponding average marginal effects of minority status on denial probabilities. Figures 1 and 2 illustrate the model-predicted denial probabilities for White and minority applicants and individual racial groups.

Table 3: Results for Main Loan Denial Logit Models

	Dependent variable: $P(\text{Denial})$		
	Model 1	Model 2	Model 3
Minority	0.485*** (0.108)		0.415*** (0.111)
Asian		0.265 (0.140)	
Black		0.844*** (0.140)	
Hispanic		0.300 (0.201)	
Non-Primary Minority		0.955 (0.528)	
ln(Income)	-0.367** (0.124)	-0.346** (0.123)	-0.364** (0.140)
LTV Ratio	0.011** (0.003)	0.009* (0.003)	0.010** (0.003)
36-43% DTI Ratio	0.004 (0.162)	-0.015 (0.163)	-0.041 (0.175)
43-50% DTI Ratio	0.299 (0.167)	0.288 (0.169)	0.422* (0.200)
> 50% DTI Ratio	3.249*** (0.232)	3.243*** (0.230)	3.492*** (0.296)
Constant	-2.417*** (0.649)	-2.354*** (0.656)	-2.497** (0.774)
Lender Fixed Effects	No	No	Yes
Pseudo R-Squared	0.186	0.190	0.253
Observations	9107	9107	8079

White and < 36% DTI Ratio serve as the omitted reference categories; standard errors clustered by lender in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

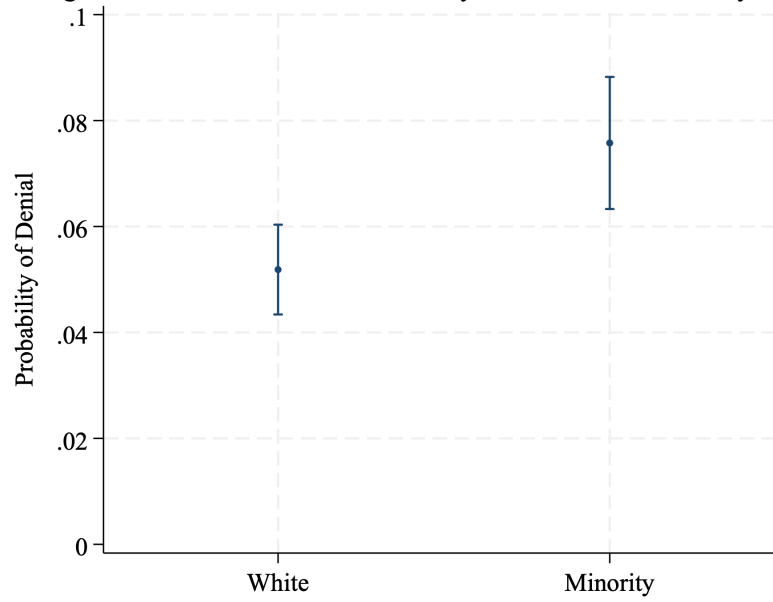
Table 4: Average Marginal Effects for Main Logit Models

	Dependent variable: $P(\text{Denial})$		
	Model 1	Model 2	Model 3
Minority	0.024*** (0.006)		0.021*** (0.006)
Asian		0.012 (0.007)	
Black		0.048*** (0.010)	
Hispanic		0.014 (0.010)	
Non-Primary Minority		0.056 (0.041)	
Observations	9107	9107	8079

White serves as the omitted reference category; standard errors clustered by lender in parentheses.

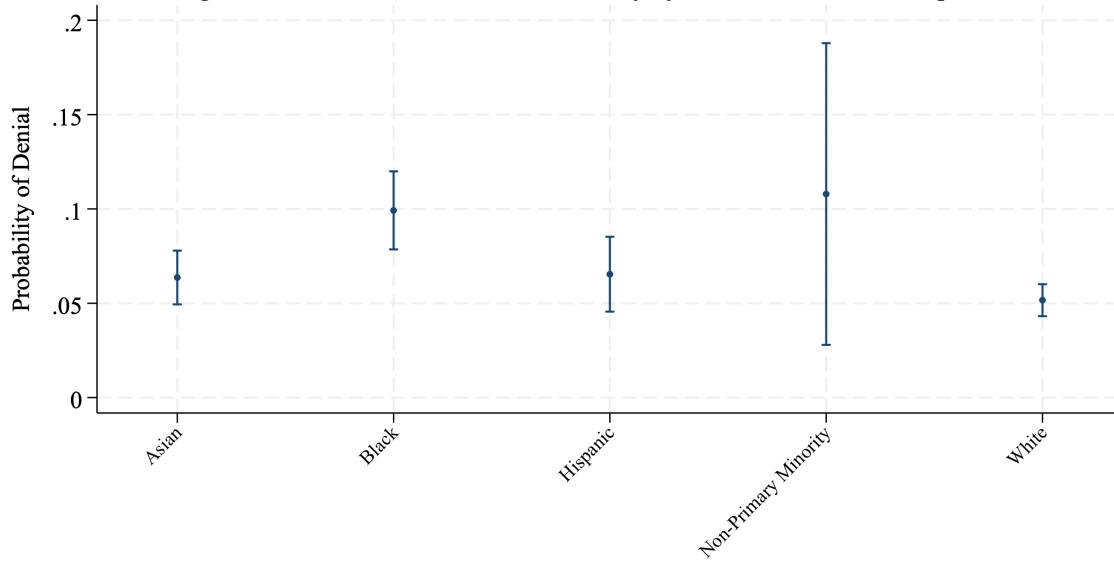
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 1: Predicted White vs. Minority Loan Denial Probability



Error bars represent a 95% confidence interval.
Controls: Income, LTV, DTI.

Figure 2: Predicted Loan Denial Probability by Racial and Ethnic Group



Error bars represent a 95% confidence interval.
Controls: Income, LTV, DTI.

The Model 1 regression results notably present a minority regressor coefficient of 0.485 that is statistically significant at the 0.1 percent level ($p < 0.001$). This provides strong evidence that, holding all else equal, minority status is associated with an increase of 0.485 in the log-odds of mortgage denial relative to a White applicant. To provide a more intuitive interpretation of this effect, Table 4 reveals that the average marginal effect of being a minority results in an absolute denial percentage increase of 2.4 percentage points compared to a comparable White individual. This estimate is also statistically significant at the 0.1 percent level. Given that the average predicted denial probability for a White applicant is approximately 5.2 percent, as shown in Figure 1, the average marginal effect of 2.4 percentage points reflects a relative increase in the probability of denial by roughly 46.1 percent.

Additionally, the coefficients on the other regressors included in the main model yield noteworthy interpretations. The coefficient for logged income is negative while the LTV ratio estimate is positive, with both being statistically significant at the 1 percent level ($p < 0.01$). These signs are econometrically intuitive, as higher income levels logically reduce the probability of denial, whereas increasing one's loan-to-value ratio should increase denial likelihood. Among the debt-to-income categorical regressors, only the indicator coefficient for a DTI ratio exceeding 50 percent is statistically significant. The coefficient is positive and large in magnitude, corresponding to a large increase in the log-odds of denial. This result is sensible as it aligns with the 50 percent DTI limit permitted by Fannie Mae and Freddie Mac for applications analyzed with their automated underwriting systems (Federal Housing Finance Agency Office of Inspector General, 2019). The statistical insignificance of the remaining DTI indicators implies that applicants in these categories do not have a meaningfully higher denial probability than those with a DTI ratio below 36 percent. A Pseudo- R^2 of 0.186 for Model 1 is near the 0.2 threshold for excellent explanatory power established by McFadden (1977), suggesting the included regressors capture a significant amount of the variance in denial outcomes.

The regression results for Model 2 indicate that differences across racial groups persist but are not uniform. Importantly, the Black indicator coefficient is the only statistically significant racial group estimator. While the coefficients for the other groups are positive, they lack the statistical power to reject the null hypothesis. The statistical insignificance of the

estimates for non-Black minorities is further evidenced in Figure 2, as the 95 percent confidence interval for Black applicants is the only interval that does not include the estimated probability for White applicants. The high variability in the Non-Primary Minority estimate is likely attributable to the small sample size of the demographic group, and thus should be interpreted with caution.

Table 4 illustrates that while all other racial marginal effects are positive yet statistically insignificant, the Black regressor is associated with a highly statistically significant value of 0.048. This figure implies that the average marginal effect of being Black results in a 4.8 percentage point increase in the probability of denial compared to a White applicant with similar attributes. Model 2 predicts that White applicants are, again, denied with a probability of 5.2 percent, so the AME of 4.8 percentage points for Black individuals reflects a considerable relative increase in the probability of denial of approximately 92.2 percent. The coefficients for the remaining non-racial variables are comparable to those of Model 1 and the specification yields a similar Pseudo- R^2 of approximately 0.190. These results suggest that Black borrowers account for the majority of disparate denial outcomes affecting the minority population in Boston.

Model 3 maintains the same regressors as Model 1 but adds lender fixed effects to its specifications to control for lender heterogeneity. While such heterogeneity is often simplified to embody differences in strictness across lenders, it also encompasses unobserved factors, such as varying levels of loan officer discretion and institutional cultures. According to Bhutta et al. (2024), excess denials at certain institutions often originate from imposing harsher standards when analyzing both observed and unobserved applicant characteristics. Model 3 controls for these differences by including a categorical variable for every lending firm in the sample. These inclusions serve two primary purposes. First, they account for the possibility that minorities may disproportionately apply to stricter lenders. Second, they allow for an evaluation of individual institutions to determine whether their strictness is applied homogeneously across racial groups. Lenders who approved or denied every applicant in 2023 were excluded from the analysis to address perfect prediction issues, resulting in a smaller sample size of 8,079 observations for Model 3.

The regressor estimate for the minority indicator in Table 3 is reduced to 0.415. With a predicted baseline denial probability of 5.9 percent for White individuals, the average marginal effect of 2.1 percentage points represents a 35 percent relative increase in the likelihood of denial for minority applicants. All of these estimates are statistically significant at the 0.1 percent level. Because Model 3 incorporates lender fixed effects, this result indicates that even when applying to the same institution, loan outcome disparities persist for comparable White and minority applicants. The remaining coefficients are consistent with the baseline results, with the exception of the indicator for a debt-to-income ratio between 43 and 50 percent, which is statistically significant at the 5 percent level ($p < 0.05$). Model 3's Pseudo- R^2 value of 0.253 represents an excellent fit. Therefore, while the results demonstrate that institutional heterogeneity accounts for a portion of the racial disparity observed in the baseline model, controlling for lender fixed effects does not completely eliminate the role of race in predicting denial outcomes.

The regression results and average marginal effects for the robustness models, found in Table 5 and Table 6, respectively, can be referenced in the Appendix. The high-income sample used in Model 4 restricted the analysis to those whose salaries exceeded \$150,000 annually, which decreased the sample size to 4,797 observations. Notably, the minority coefficient decreased to 0.420 and is only statistically significant at the 5 percent level. Additionally, the estimate for the logged income regressor is still positive but not statistically significant. The predicted average marginal effect for minority status also attenuates to 1.5 percentage points. A Pseudo- R^2 of 0.140 represents a modest fit. These results indicate that while still relevant, race and income have a decreased affect on the probability of denial for high-earning individuals.

Model 5 accounts for neighborhood differences, mirroring the methodology of Munnell et al. (1996). The specification includes the Tract-to-MSA Income percentage, which controls for the possibility that lenders are less willing to approve mortgage loans for applicants purchasing properties in lower-income neighborhoods. The coefficient on the Tract-to-MSA Income percentage variable is near zero and statistically insignificant, indicating that neighborhood income plays a minimal role in predicting denial outcomes. However, its inclusion reduces the minority coefficient to 0.454 relative to the baseline model's estimate. The average marginal effect for minority applicants also slightly declines to 0.022, representing an absolute denial

increase of 2.2 percentage points compared with an otherwise similar White applicant. The model exhibits a Pseudo- R^2 of 0.194. Therefore, while neighborhood income explains part of the disparities in the probability of denial credited to minority status in the baseline, the effect of race still persists.

The final two models control for aspects of automated underwriting practices. While Bhutta et al. (2024) centered a substantial portion of their analysis on differences between colorblind AUS recommendations and the loan officer's ultimate application decision, such evaluation is not possible in this paper. As noted previously, information on the suggestion of the AUS is only available in the confidential HMDA extract, which is not accessible to the public. Consequently, Model 6 chooses to analyze whether reporting AUS usage is associated with differences in denial outcomes.

In this specification, an additional omitted reference category is included for applications where AUS usage missing or exempt.. The coefficient for the reported institutional use of an automated underwriting system is -0.612, which is statistically significant at the 1 percent level. Such a figure indicates that reported AUS usage decreases the probability of denial. Crucially, the minority regressor and marginal effect estimates of 0.471 and 0.023 remain largely stable. Both values are statistically significant at the 0.1 percent level and represent minor decreases from the baseline. Model 6 has a Pseudo- R^2 value of 0.197. These results suggest that the reported use of automated underwriting systems may be associated with more lenient approval outcomes relative to manual or unreported practices, yet it is not a primary source of racial disparities.

Model 7 examines whether the specific automated underwriting algorithm used during the loan evaluation process helps explain observed denial differences. Due to the way in which the specific algorithm indicators were defined, their corresponding coefficients represent the association between the presence of a given system type and denial outcomes, conditional on the other systems reported. The regressor estimate for a GSE-originated system is -0.590 and is statistically significant at the 1 percent level. Conversely, the indicator for an internal proprietary AUS is 1.140, a large positive coefficient that is statistically significant at the 0.1 percent level. This contrast provides evidence of the presence of harsher institutional 'overlays' as described by

Bhutta et al. (2024). A 0.203 Pseudo- R^2 value represents an excellent fit for the model. Nevertheless, with minority estimates that are comparable to Model 6, controlling for the particular automated underwriting system employed does not bridge the gulf in outcome predictions on the basis of race.

Finally, to ensure the validity of the main logistic model, a Link Test was performed for model specification. The resulting coefficient for the squared latent variable, $\hat{z}^2$, is statistically insignificant with a p-value of 0.758. This suggests that Model 1 is correctly specified and does not suffer from significant biases in the underlying mathematical structure that would undermine the reliability of the observed estimates.

6. CONCLUSION

This study aimed to determine whether the disparate mortgage loan denial gaps first identified by Munnell et al. (1996) persist in Boston in 2023. The evidence suggests that after controlling for borrower characteristics, loan attributes, lender fixed effects, a high-income sample, neighborhood differences, and reported automated underwriting systems practices, such denial gaps remain. Furthermore, the models demonstrate that these outcome differences are partially attributable to racial identity. Black individuals appear to account for a large portion of the disparities.

However, it is important to state that the scope of these findings is limited. Because the public HMDA extract lacks credit score information, the included models cannot control for default risk, potentially introducing omitted variable bias. Without access to the confidential 2023 dataset, it is impossible to ascertain whether the inclusion of credit measures would shrink the observed denial gap or render racial identity statistically insignificant. These findings should prompt further investigation of post-COVID mortgage applications to definitively account for credit worthiness. Ultimately, despite data availability restraints, the included models' consistent results provide evidence that the racial disparities documented by Munnell et al. (1996) remain today notwithstanding decades of changes to underwriting practices and standards.

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APPENDIX

Table 5: Results for Robustness Loan Denial Logit Models

	Dependent variable: $P(\text{Denial})$			
	Model 4	Model 5	Model 6	Model 7
Minority	0.420* (0.180)	0.454*** (0.109)	0.471*** (0.105)	0.473*** (0.103)
ln(Income)	0.056 (0.240)	-0.328* (0.129)	-0.434*** (0.120)	-0.477*** (0.125)
LTV Ratio	0.006 (0.007)	0.010** (0.004)	0.012** (0.004)	0.012*** (0.004)
36-43% DTI Ratio	0.263 (0.178)	0.016 (0.167)	0.001 (0.160)	-0.012 (0.159)
43-50% DTI Ratio	0.814*** (0.188)	0.311 (0.172)	0.345* (0.167)	0.358* (0.168)
> 50% DTI Ratio	3.896*** (0.229)	3.269*** (0.240)	3.242*** (0.226)	3.257*** (0.226)
Tract-to-MSA Income		-0.002 (0.001)		
AUS Usage Reported			-0.612** (0.212)	
GSE AUS Used				-0.590** (0.209)
Internal AUS Used				1.140*** (0.202)
Other AUS Used				0.038 (0.333)
Constant	-4.592** (1.471)	-2.375*** (0.653)	-1.648* (0.700)	-1.564* (0.704)
Pseudo R-Squared	0.140	0.194	0.197	0.203
Observations	4797	9107	9107	9107

White and < 36% DTI Ratio serve as the omitted reference categories for Models 4-7; Tract-to-MSA Income is measured as a percentage of MSA Income; Exempt/NA AUS Usage serves as an additional omitted reference category for Models 6 and 7; standard errors clustered by lender in parentheses.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6: Average Marginal Effects for Robustness Logit Models

	Dependent variable: $P(\text{Denial})$			
	Model 4	Model 5	Model 6	Model 7
Minority	0.015* (0.007)	0.022*** (0.006)	0.023*** (0.005)	0.023*** (0.005)
Observations	4797	9107	9107	9107

White serves as the omitted reference category; standard errors clustered by lender in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$